

39. SUMMARY OF MOVEMENT

DURATION : 05:02

STUDY SHEET

COURSE SUMMARY

This video delves into the heart of embryonic movement, detailing the crucial role of the electrical and electromagnetic axis in establishing a differential growth rate. You will learn how the notochordal process initiates cerebralisation, influencing the development of essential structures such as the digestive tract and the cardiac system. The concepts of palpation and the interconnection between different embryonic layers are explored to better understand the dynamics between the heart, brain, and other organs. Enjoy this enriching synthesis that highlights the complex interactions that shape the embryo.

SUMMARY OF MOVEMENT

ESTABLISHMENT OF THE AXIS AND DIFFERENTIAL GROWTH

The initial process involves the establishment of an **electrical** and **electromagnetic axis**. This axis progressively organises a **differential growth rate**.

This growth rate acts as an **algorithm** integrated into the system, causing at a given moment the precise **"hola"** (wave) of the **notochord**.

NOTOCHORDAL AND MESODERMAL PROCESS

The **notochordal** process forms and simultaneously allows for the establishment of the **lateral mesodermal** process. At the same instant, **symmetry** begins to establish itself.

A **fulcrum** emerges at the base of the skull, precisely at the level of the future **sphenobasilar synchondrosis**. This fulcrum is supported by the developmental potential of the **sacrum**.

CEREBRALISATION AND CONCOMITANT DEVELOPMENTS

This development marks the beginning of **cerebralisation**. Cerebralisation induces **cardialisation**, which in turn leads to **diaphragmatisation**, then **hepatisation**.

Posterior growth causes **pneumatisation**, then posterior **adrenalisation**, and so on.

CEREBRAL EVOLUTION AND KEY POINTS

During the **cephalic growth** phase, the **foramen of Monro** opens laterally, clearing the **ventricular** system. This process accompanies cerebralisation and **corticalisation**.

A posterior **neurulation** reappears, relying on a key point called the **Insula**.

The fundamental point of stillness remains the tip of the notochord. However, in the brain, the **Insula** constitutes the fulcrum and the crucial tipping point for development.

ASCENDING AND DESCENDING MOVEMENTS

A process of **ascension** and a process of **descent** organise the entire development. These two movements are interdependent.

When cerebralisation occurs, synchronous events take place:

- The development of the **digestive tube**
- The development of the **cardiac tube**
- The development of the **pulmonary tube**

This ascent of the tube and this descent are crucial. The most important area that allows these phenomena is that of the **heart** and its relationship with the mother.

NOTOCHORD, NEURAL PLATE AND PRECARDIAC ZONE

The notochord induces the formation of the **neural plate** at the anterior of the embryo. Two **fluidic trajectories** are created.

These trajectories coil upwards to form the **precardiac zone**.

DEPTHS OF PALPATION AND HITCH POINTS

There are different **depths of palpation** to consider:

1. **Palpation of the Hitchpoint (fulcrum):** Linked to brain development.
2. **Palpation of the brain:** In relation to its position and developmental dynamics.
3. **Pure development:** Concerning relationships with the heart, liver, and digestive system.

In terms of depth, several layers can be identified:

- **Superficial zone:** Contact with hair, then skin and bone.
- **First layer:** The **visceral dural layer**, just behind the bone.
- **Fluid layer** protecting the brain.
- **Cortical layer:** To feel its movement.

- **The nuclei:** Descending deeper.
- **The ventricular plane:** Mainly lateral.